2,2-DIFLUOROPROPIONAMIDE DERIVATIVES OF BARDOXOLONE METHYL, POLYMORPHIC FORMS AND METHODS OF USE THEREOF

The present application is a continuation application of pending U.S. patent application Ser. No. 14/625,829, filed Feb. 19, 2015, which is a continuation of U.S. patent application Ser. No. 13/869,833, filed Apr. 24, 2013, now U.S. Pat. No. 8,993,640, which claims the benefit of priority to U.S. Provisional Application No. 61/780,444, filed on Mar. 13, 2013, U.S. Provisional Application No. 61/775,288, filed on Mar. 8, 2013, and U.S. Provisional Application No. 61/687,669, filed on Apr. 27, 2012; the entire contents of each are herein incorporated by reference.

Pursuant to 37 C.F.R. 1.821(c), a sequence listing is submitted herewith as an ASCII compliant text file named "REATP0073USC2_ST25," created on Jun. 1, 2017 and having a size of ~7 kilobytes. The content of the aforementioned file is hereby incorporated by reference in its entirety.

BACKGROUND OF THE INVENTION

I. Field of the Invention

The present invention relates generally to the compound: N-((4aS,6aR,6bS,8aR,12aS,14aR,14bS)-11-cyano-2,2,6a,6b,9,9,12a-heptamethyl-10,14-dioxo-1,2,3,4,4a,5,6,6a,6b,7,8,8a,9,10,12a,14,14a,14b-octadecahydropicen-4a-yl)-2,2-difluoropropanamide,

also referred to herein as RTA 408, 63415, or PP415. The present invention also relates to polymorphic forms thereof, methods for preparation and use thereof, pharmaceutical compositions thereof, and kits and articles of manufacture thereof.

II. Description of Related Art

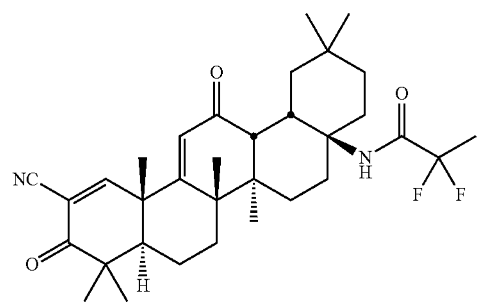
The anti-inflammatory and anti-proliferative activity of the naturally occurring triterpenoid, oleanolic acid, has been improved by chemical modifications. For example, 2-cyano-3,12-diooxooleana-1,9(11)-dien-28-oic acid (CDDO) and related compounds have been developed. See Honda et al., 1997; Honda et al., 1998; Honda et al., 1999; Honda et al., 2000a; Honda et al., 2000b; Honda et al., 2002; Suh et al., 1998; Suh et al., 1999; Place et al., 2003; Liby et al., 2005; and U.S. Pat. Nos. 8,129,429, 7,915,402, 8,124,799, and 7,943,778, all of which are incorporated herein by reference. The methyl ester, bardoxolone methyl (CDDO-Me), has been evaluated in phase II and III clinical trials for the treatment and prevention of diabetic nephropathy and chronic kidney disease. See Pergola et al., 2011, which is incorporated herein by reference.

Synthetic triterpenoid analogs of oleanolic acid have also been shown to be inhibitors of cellular inflammatory processes, such as the induction by IFN-γ of inducible nitric oxide synthase (iNOS) and of COX-2 in mouse macrophages. See Honda et al, (2000a), Honda et al. (2000b), Honda et al. (2002), and U.S. Pat. Nos. 8,129,429, 7,915,402, 8,124,799, and 7,943,778, which are all incorporated herein by reference. Compounds derived from oleanolic acid have been shown to affect the function of multiple protein targets and thereby modulate the activity of several important cellular signaling pathways related to oxidative stress, cell cycle control, and inflammation (e.g., Dinkova-Kostova et al., 2005; Ahmad et al., 2006; Ahmad et al., 2008; Liby et al., 2007a, and U.S. Pat. Nos. 8,129,429, 7,915,402, 8,124,799, and 7,943,778).

Given that the biological activity profiles of known triterpenoid derivatives vary, and in view of the wide variety of diseases that may be treated or prevented with compounds having potent antioxidant and anti-inflammatory effects, and the high degree of unmet medical need represented within this variety of diseases, it is desirable to synthesize new compounds with different biological activity profiles for the treatment or prevention of one or more indications.

SUMMARY OF THE INVENTION

In some aspects of the present invention, there is provided a compound of the formula (also referred to herein as RTA 408, 63415, or PP415):



or a pharmaceutically acceptable salt thereof.

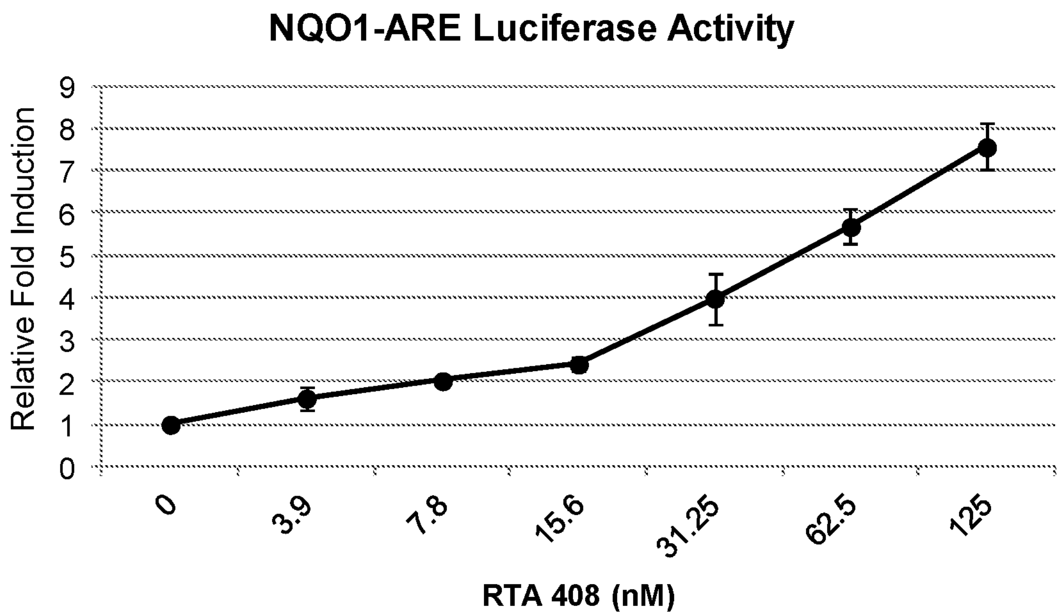
In some embodiments, the compound is in the form of a pharmaceutically acceptable salt. In some embodiments, the compound is not in the form of a salt.

In another aspect of the present invention, there are provided polymorphic forms of the above compound. In some embodiments, the polymorphic form has an X-ray powder diffraction pattern (CuKα) comprising a halo peak at about 14 °2θ. In some embodiments, the X-ray powder diffraction pattern (CuKα) further comprises a shoulder peak at about 8 °2θ. In some embodiments, the X-ray powder diffraction pattern (CuKα) is substantially as shown in FIG. **59**. In some embodiments, the polymorphic form has a T_g from about 150° C. to about 155° C., including for example, a T_g of about 153° C. or a T_g of about 150° C. In some embodiments, the polymorphic form has a differential scanning calorimetry (DSC) curve comprising an endotherm centered from about 150° C. to about 155° C. In some embodiments, the endotherm is centered at about 153° C. In some embodiments, the endotherm is centered at about 150° C. In some embodiments, the differential scanning calorimetry (DSC) curve is substantially as shown in FIG. **62**.

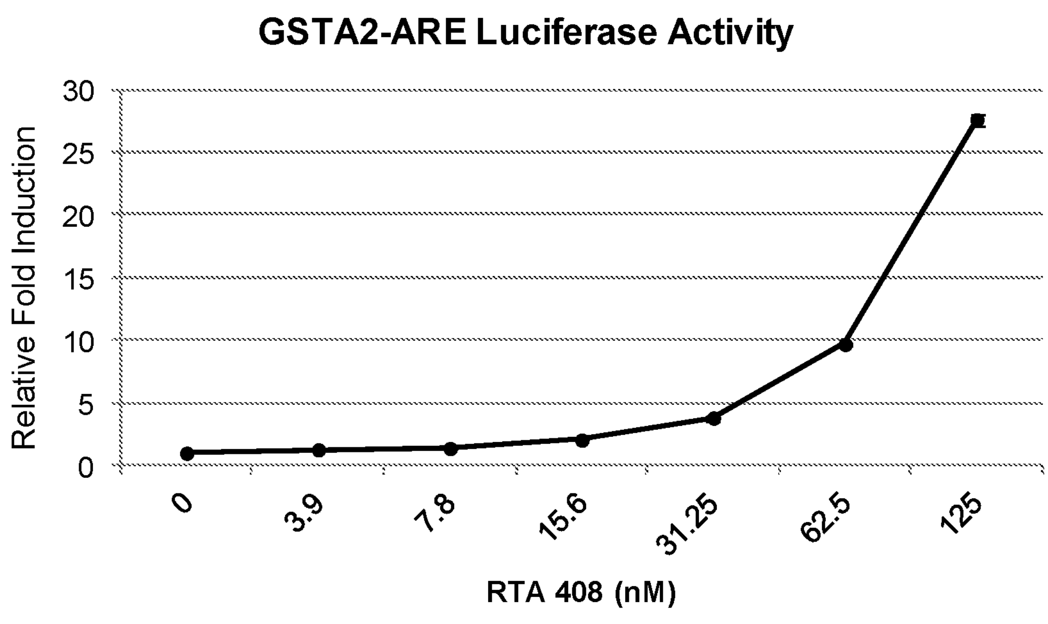
In some embodiments, the polymorphic form is a solvate having an X-ray powder diffraction pattern (CuKα) comprising peaks at about 5.6, 7.0, 10.6, 12.7, and 14.6 °2θ. In some embodiments, the X-ray powder diffraction pattern (CuKα) is substantially as shown in FIG. **75**, top pattern.

In some embodiments, the polymorphic form is a solvate having an X-ray powder diffraction pattern (CuKα) comprising peaks at about 7.0, 7.8, 8.6, 11.9, 13.9 (double peak), 14.2, and 16.0 °2θ. In some embodiments, the X-ray diffraction pattern (CuKα) is substantially as shown in FIG. **75**, second pattern from top.

In some embodiments, the polymorphic form is an acetonitrile hemisolvate having an X-ray powder diffraction


NQO1-ARE Luciferase Activity
Relative Fold Induction
0
1
2
3
4
5
6
7
8
9
0
3.9
7.8
15.6
31.25
62.5
125
RTA 408 (nM)

(a)


GSTA2-ARE Luciferase Activity
Relative Fold Induction
0
5
10
15
20
25
30
0
3.9
7.8
15.6
31.25
62.5
125
RTA 408 (nM)

(b)

FIGS. 2a & b